Characterisation of Dynkin and Euclidean graphs

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A. Direct Sum Decompositions

Let A be a finite-dimensional $K-$algebra (throughout the poster).

- Idempotents $e_1, e_2 \in A$ are called **orthogonal** if $e_1e_2 = e_2e_1 = 0$.
- Idempotent e is called **primitive** if e cannot be written as a sum $e = e_1 + e_2$, where e_1 and e_2 are non-zero orthogonal idempotents of A.
- The left $A-$module ${}_AA$ admits a direct sum decomposition:

$$ {}_AA = P_1 \oplus ... \oplus P_n$$

where P_1, ..., P_n are indecomposible left ideals with $P_i = Ae_i$, $\forall i = 1, ..., n$, where $e_1, ..., e_n$ are primitive pairwise orthogonal idempotents of A such that $1 = e_1 + ... + e_n$.

– The above is called an **indecomposable decomposition** of A.

– $\{e_1, ..., e_n\}$: complete set of **primitive orthogonal idempotents** of A.

B. Dimension vector of a module

Let M be a module in $A-$mod. The **dimension vector** of M is:

$$\text{dim}M = \begin{bmatrix} \text{dim}_K e_1 M \\ . \\ . \\ . \\ \text{dim}_K e_n M \end{bmatrix} = \begin{bmatrix} \text{dim}_K \text{Hom}(P(1), M) \\ . \\ . \\ . \\ \text{dim}_K \text{Hom}(P(n), M) \end{bmatrix}$$

where $e_1, ..., e_n$ are primitive orthogonal idempotents of A.

- Throughout, by dim$M$, we refer to the dimension vector with respect to a given complete set $\{e_1, ..., e_n\}$ of primitive orthogonal idempotents of A.
- By unique decomposition, dim$M$ doesn't depend on the choice of the complete set, up to permutation of its coordinates.

C. Grothendieck Group of A

The **Grothendieck group**, $K_0(\mathscr{A})$ of an abelian category, $\mathscr{A}$, is the abelian group generated by the symbols, $[P]$, where $P \in \text{Ob}(\mathscr{A})$, subject to the relations: $[P_2] = [P_1] + [P_3]$, when $0 \to P_1 \to P_2 \to P_3 \to 0$ is a short exact sequence.

- If $P_1 \cong P_2$, then $[P_1] = [P_2]$
- $K_0(A-\text{mod}) \cong \mathbb{Z}^n$

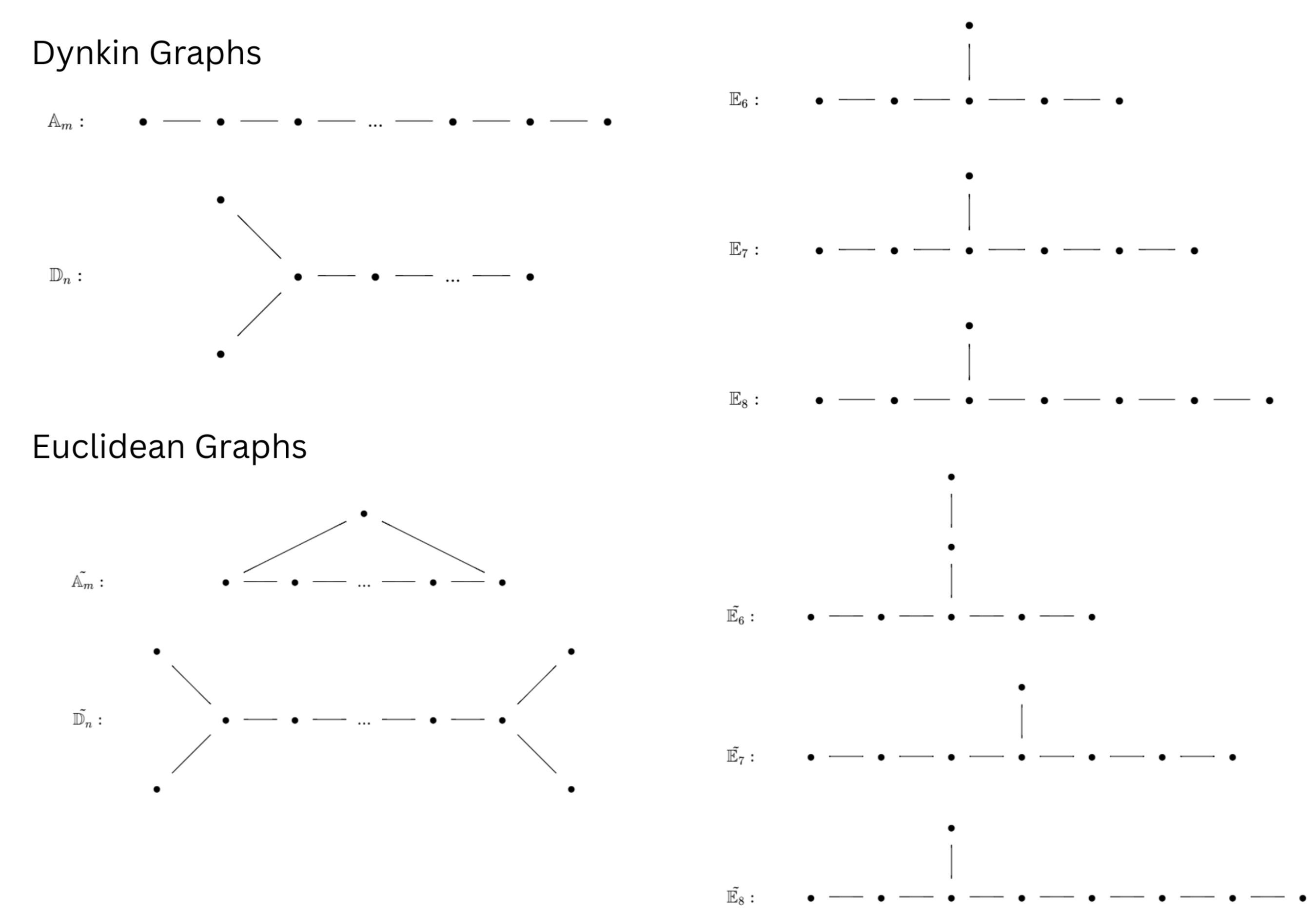
- $K_0(A-\text{mod})$ is a free abelian group having $\{[S(1)], ..., [S(n)]\}$ as a basis.
- When $A \cong KQ$, for a quiver, $Q = (Q_0, Q_1, s, t)$, each $S(i)$ corresponds to a vertex i in Q_0.

D. Cartan Matrix

Given A and $\{e_1, ..., e_n\}$ as before, the **Cartan Matrix** is $C_A = [c_{ij}]_{n\times n}$, where $c_{ij} = e_iAe_j$ for $i, j = 1, ..., n$.

- The ith column of C_A is dim$P(i)$.
- The ith row of C_A is $[\text{dim}I(i)]^t$.
- C_A is invertible with det $C_A \in \{-1, 1\}$.

Characterisation of Dynkin and Euclidean Graphs using Quadratic forms



Lemma 1 Let Q be a finite connected acyclic quiver. If the underlying graph $\bar{Q}$ is not a Dynkin graph, then $\bar{Q}$ contains a Euclidean graph as a subgraph.
– Proof by contrapositive. Eliminating each of the Euclidean graphs gives an underlying graph that coincides with one of the Dynkin graphs.

Lemma 2 Let Q be a quiver whose underlying graph $\bar{Q}$ is Euclidean. Then q_Q is positive semidefinite of corank one.
– Follows from the quadratic forms of the Euclidean graphs.

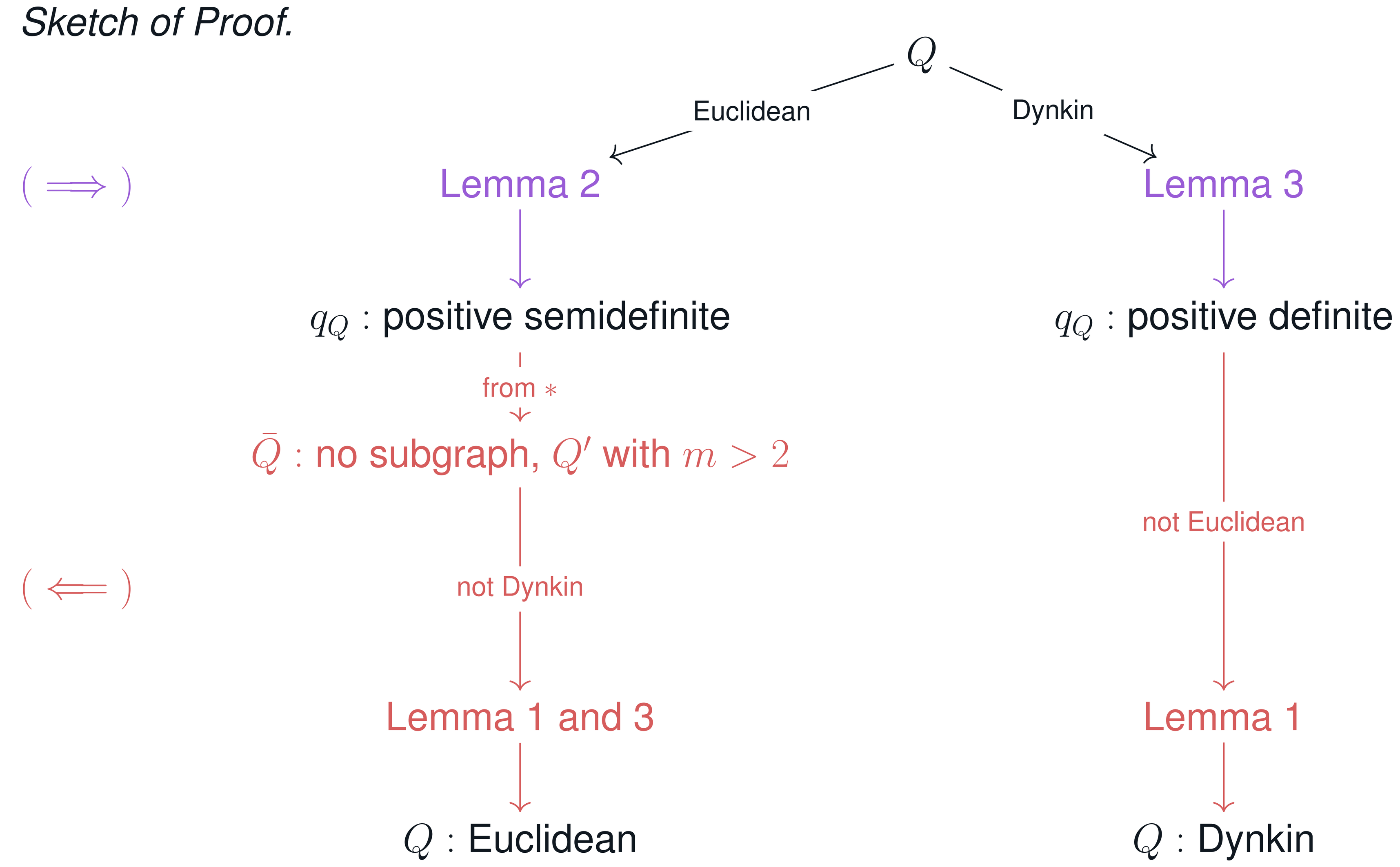
Lemma 3 Let Q be a connected quiver with q_Q positive semidefinite and Q', a proper subquiver of Q. Then the restriction of q_Q to Q' is positive definite.

Cor. For a quiver Q with $\bar{Q}$ as Dynkin, q_Q is positive definite.

Theorem For a finite connected acyclic quiver Q, with underlying graph $\bar{Q}$

1. $\bar{Q}$ is a Dynkin graph $\iff$ q_Q is positive definite.
2. $\bar{Q}$ is a Euclidean graph $\iff$ q_Q is positive semidefinite but not positive definite.

Sketch of Proof.



E. Euler characteristic & Quadratic form

- The **Euler characteristic** of A is the bilinear (non-symmetric) form $\langle -, - \rangle_A : \mathbb{Z}^n \times \mathbb{Z}^n \to \mathbb{Z}$ defined by $\langle x, y \rangle_A = x^t (C_A^{-1})^t y$.
- The **Euler quadratic form** of an algebra, A, is $q_A : \mathbb{Z}^n \to \mathbb{Z}$ defined by $q_A(x) = \langle x, x \rangle_A$.

F. Integral Quadratic forms

A quadratic form, $q = q(x_1, ..., x_n)$ on $\mathbb{Z}^n$ in n indeterminates, $x_1, ..., x_n$ is said to be an **integral quadratic form** if it is of the form, $q(x_1, ..., x_n) = \sum_{i=1}^n x_i^2 + \sum_{i<j} a_{ij}x_ix_j$, where $a_{ij} \in \mathbb{Z}\ \forall i, j$.

- Evaluating such a form, q, on $x = [x_1, ..., x_n]^t \in \mathbb{Z}^n$ gives us a mapping from $\mathbb{Z}^n$ to $\mathbb{Z}$, denoted by q.
  - $x \in \mathbb{Z}^n$ is called **positive**, if $x \neq 0$ and $x_i \geq 0\ \forall\ 1 \leq i \leq n$.
  - q is called **positive semidefinite** if $q(x) \geq 0$ for all $x \in \mathbb{Z}^n$.
  - q is called **positive definite** if $q(x) > 0$ for all $x \in \mathbb{Z}^n \setminus \{0\}$.
  - q is called **indefinite** if $\exists x \in \mathbb{Z}^n \setminus \{0\}$ such that $q(x) < 0$.
- For a positive semidefinite form, q, the **radical** of q is:

$$\text{rad } q = \{x \in \mathbb{Z}^n | q(x) = 0\}$$

  - rad q is a subgroup of $\mathbb{Z}^n$. Its rank is called the **corank** of q.
  - q is positive definite iff its corank is 0.

G. Quadratic form of a quiver

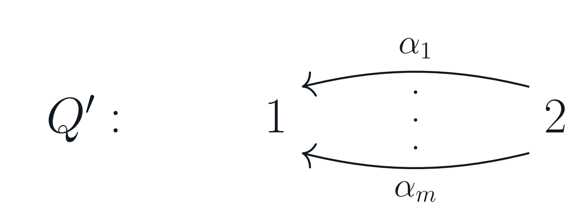
The **quadratic form** of a quiver Q is

$$q_Q(x) = \sum_{i \in Q_0} x_i^2 - \sum_{\alpha \in Q_1} x_{s(\alpha)}x_{t(\alpha)}$$

where $x = [x_1, ..., x_n]^t \in \mathbb{Z}^n$.

- q_Q depends on the underlying graph, not the arrow directions.
- For a finite connected acyclic quiver, Q, the Euler quadratic form of q_{KQ} coincides with the quadratic form of the quiver, q_Q.

Example $(*)$:



$$q_{Q'}(x) = x_1^2 + x_2^2 - mx_1x_2 = (x_1 - \tfrac{m}{2}x_2)^2 + (1 - \tfrac{m^2}{4})x_2^2$$

$\implies$ $q_{Q'}$ is positive definite if $m = 1$, positive semidefinite of corank 1 if $m = 2$, and indefinite otherwise.

References

[1] I. Assem, D. Simson, and A. Skowronski. *Elements of the Representation Theory of Associative Algebras: Volume 1: Techniques of Representation Theory.* Cambridge University Press, 2006. ISBN: 9780521584234.